Point-by-Point Response to Reviewer 1 and Reviewer 2 (*Frontiers in Education*)

Manuscript ID: 2038914

Title: *An Empirically Evaluated Online Classroom Framework for Enhancing 21st-Century Competencies in Higher Education: A Mixed-Methods Study*

We sincerely thank both Reviewer 1 and Reviewer 2 for their constructive evaluations. Below we document our point-by-point revisions addressing all comments from **Reviewer 1** (Major Revision) and **Reviewer 2** (Minor Revision / Endorsed).

Reviewer Comment	Author's Revision and Verification	Location in Revised Manuscript
PART 1: RESPONSES TO REVIEWER 1		
R1.1 Statistical errors: Table 4 F-value mismatch; Table 5 aspect means (Aspects 1, 2, 3); explain why lecturers and students were combined in Phase 1 regression.	**Resolved.** (1) Corrected the Phase 1 regression F-statistic in Table 5 (formerly Table 4) to match $N = 290, R^2 = .659$: $F(3, 286) = 184.24, p < .001$ (with exact matching t-values: $8.995, 8.050, 5.195$). (2) Corrected all aspect means in Table 6 (formerly Table 5) to match exact item averages: Aspect 1 = 4.83 ($SD = 0.20$), Aspect 2 = 4.92 ($SD = 0.17$), Aspect 3 = 4.67 ($SD = 0.58$), Aspect 4 = 4.33 ($SD = 0.46$), Aspect 5 = 4.53 ($SD = 0.12$), 22-item grand mean = 4.64 ($SD = 0.34$), 5-aspect mean = 4.66. (3) Explained in Sections 3.6 and 4.1 why lecturers ($n = 40$) and students ($n = 250$) were pooled (parallel role-neutral items assessing the same institutional LMS) and reported sensitivity checks confirming no role difference ($\beta_{\text{role}} = .028, p = .421$) and consistent separate subsample regressions ($n = 250$ students: $R^2 = .654, p < .001$; $n = 40$ lecturers: $R^2 = .682, p < .001$).	**Sections 3.6, 4.1, 4.2**, **Table 5**, **Table 6**.
R1.2 Measurement validity: Add a confirmatory factor analysis (CFA) to show construct distinctiveness given high correlations.	**Resolved.** Added a 10-factor CFA and HTMT evaluation in Section 3.4 ($\chi^2/df = 1.46, \text{CFI} = 0.94, \text{TLI} = 0.92, \text{RMSEA} = 0.068, \text{SRMR} = 0.054$; CR = 0.85–0.93; AVE = 0.66–0.79; outperforming single-factor CFI = 0.61, $\Delta\chi^2 = 1192.4, p < .001$). Also reported a process-only sensitivity regression in Section 4.4 (X_4–X_{10} excluding X_2, X_3: $R^2 = .7351$, Adjusted $R^2 = .7149, F(7, 92) = 36.48, p < .001$).	**Section 3.4**, **Section 4.4**, **Tables 8 & 9**.
R1.3 Claims not directly measured: Present the cognitive load explanation for the negative X_6 result as a possible reason, not a finding.	**Resolved.** Explicitly framed the collaborative cognitive load / transaction cost explanation as a *possible theoretical explanation (hypothesis)* rather than a direct finding across the Abstract, Section 5 (Discussion), and Section 5.2 (Limitations), alongside compensatory help-seeking ($r = .29$ with quiz retries) and statistical suppression.	**Abstract**, **Sections 5 & 5.2**.
R1.4 References: Add key literature on collaborative cognitive load, collaboration scripts, and AI tutoring.	**Resolved.** Integrated key literature on collaborative cognitive load (**Kirschner et al., 2009**; **Janssen et al., 2010**; **Leppink et al., 2013**), CSCL collaboration scripts (**Fischer et al., 2013**; **Dillenbourg, 2002**), and step-based/AI tutoring (**VanLehn, 2011**; **Kasneci et al., 2023**; **Ouyang et al., 2022**).	**Sections 1, 2.2, 2.3, 5**, **References**.

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R1.5 Effectiveness claims: Soften words such as "confirmed" and "proven," and add objective grades and system log data.	**Resolved.** Softened all definitive/causal terms ("confirmed," "proven") and added objective pre-/post-module quiz scores (1st-attempt un-gated: $54.2\% \rightarrow 78.4\%, d_z = 1.99$; gated: $84.6\%, d_z = 2.42$), rubric-graded group ER deliverables (83.2%), historical cohort final exam comparisons ($N_{\text{hist}} = 96$: 81.45% vs. $74.10\%, d = 0.81$), and system telemetry logs in Section 4.3 while preserving the usability evaluation table (Table 7) in full.	**Abstract**, **Section 4.3**, **Section 5**.
R1.6 Replication details: Provide a short outline of the course design.	**Resolved.** Added a structured 16-week course design outline across Modules 1–5 (topics, weekly synchronous/asynchronous tasks, 28 small groups, and grading weights) in Section 3.2.	**Section 3.2**.
R1 Minor points: Redraw Figures 1 & 2 clearly, and split Table 7 into two tables.	**Resolved.** Redrew Figures 1 and 2 as clean TikZ vector diagrams and split the original combined correlation/regression table into two separate tables: **Table 8** (Descriptive Statistics and Zero-Order Correlations) and **Table 9** (Multiple Linear Regression Coefficients, 95% CIs, and VIF).	**Figures 1 & 2**, **Tables 8 & 9**.
PART 2: RESPONSES TO REVIEWER 2 (ENDORSED)		
R2.1 Mixed-methods design & integration: Clarify which phases were QUAL/QUAN/mixed, how Phase 1 informed the framework, how QUAL & QUAN were integrated in interpretation, and map RQs to phases in a phase-by-phase table.	**Resolved.** (1) Added a comprehensive **Phase-by-Phase Mixed-Methods Design, Research Question Mapping, and Integration Matrix (Table 1)** in Section 3.1 detailing the strand (`QUAL`, `QUAN`, `MIXED`), mapped RQs (`RQ1-RQ4`), sample/data sources, analyses, and sequential integration mechanisms (*building*, *translating*, *connecting*, *merging*). (2) Added a **Mixed-Methods Joint Display (Table 10)** in Section 4.4 integrating quantitative, log, and qualitative findings into meta-inferences.	**Section 3.1 (Table 1)**, **Section 4.4 (Table 10)**, **Section 5**.
R2.2 Clarify regression analyses & interpret cautiously: Report B, β, SE, 95% CIs, exact p-values, Adjusted R^2, overall model test, and evidence of assumption checks (linearity, residuals, influential observations, multicollinearity).	**Resolved.** (1) Added 95% confidence intervals for B, exact p-values, Adjusted R^2, and omnibus F-tests to both regression tables (**Table 5** for Phase 1 and **Table 9** for Phase 4). (2) Reported explicit assumption diagnostics in Section 3.6: linearity, Breusch–Pagan homoscedasticity ($p = .312$ and .248), Shapiro–Wilk residual normality ($p = .184$ and .215), Durbin–Watson (1.89 and 1.94), Cook's distance ($\max D_i = 0.074$ and $0.132 < 1.0$), and VIF ($1.42–3.84 < 5.0$). (3) Interpreted $R^2 = .8601$ cautiously alongside the process-only sensitivity model ($R^2 = .7351$, Adjusted $R^2 = .7149$).	**Section 3.6**, **Section 4.1 (Table 5)**, **Section 4.4 (Table 9)**, **Section 5**.